

Transfer Manifest - Technology #13

Self-Referencing Piezoresistive Pressure Sensor

YOU RECEIVE:

- + Self-referencing sensor concept + R1 correction (drift compensation architecture)
- + Piezoresistive + Wheatstone bridge governing equations
- + Bench test protocols (calibration, drift, hermeticity, kink, EMC)
- + External precedent catalog (MEMS pressure sensors, drift compensation)
- + Critical UNKNOWN disclosure (achievable drift, hermeticity, sensor survival)
- + This engineering dossier with failure analysis + build plan

YOU MUST DEVELOP:

- > Production MEMS sensor design (diaphragm, piezoresistors, geometry)
- > Self-referencing bridge architecture + signal conditioning
- > Hermetic packaging process
- > Catheter integration manufacturing
- > Regulatory submission (active implantable, ISO 14708-1, IEC 60601-1-2)
- > Clinical validation (accuracy + drift over implantation duration)
- > Long-term reliability testing

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

NOT AVAILABLE:

- Physical prototype
- Clinical data
- Qualified supplier
- Granted IP
- Independent engineer evaluation

This manifest is honest. Items marked 'not available' do not exist.